Elementary Analysis

Section 37: A Discussion of Exponents and Logarithms

Exercises 37.1–37.9, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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37 A discussion of exponents and logarithms

In this section we must avoid using familiar exponential and logarithm rules before they have been proved. Our starting point is

$$L(y) = \int_1^y \frac{dt}{t} \quad (y > 0).$$

Theorem 37.4 proves that $L'(y) = 1/y$, that L is strictly increasing onto $\mathbb{R}$, and that $L(yz) = L(y) + L(z)$. Its inverse is E, so $L(E(x)) = x$ and $E(L(y)) = y$. Theorem 37.6 proves $E' = E$ and $E(u + v) = E(u)E(v)$. Real powers are then *defined* by

$$b^x = E(xL(b)) \quad (b > 0,\ x \in \mathbb{R}).$$

These are the integral-based results permitted by the exercise instructions. No result from the informal approaches in 37.1 or 37.2 is needed below. We write $\log_e y = L(y)$ after that identification has been established.

37.1 Exercise 37.1: Prove the product law directly from the integral

What we must show. For $y, z > 0$, show directly that $\int_1^{yz} dt/t = \int_1^y dt/t + \int_1^z dt/t$.

Plan. Split at y and rescale the remaining interval by $t = yu$.

Use additivity with oriented endpoints.

All endpoints are positive, so $1/t$ is continuous on the compact interval containing $1, y, yz$. The oriented-integral convention gives

$$\int_1^{yz} \frac{dt}{t} = \int_1^y \frac{dt}{t} + \int_y^{yz} \frac{dt}{t}.$$

This equality remains true even when $y < 1$ or $yz < y$; reversing endpoints changes the sign of an integral.

Make the substitution, including its scale factor.

Set $t = yu$, so $dt = y\,du$. The lower endpoint $t = y$ becomes $u = 1$, and $t = yz$ becomes $u = z$. Therefore

$$\int_y^{yz} \frac{dt}{t} = \int_1^z \frac{y\,du}{yu} = \int_1^z \frac{du}{u}.$$

Because $y > 0$, this is a valid continuously differentiable change of variable, with oriented endpoints also handling $z < 1$.

Conclusion. Substituting this last identity into the split integral gives the desired formula, without assuming a logarithm product rule.

37.2 Exercise 37.2: The limits of the inverse logarithm

What we must show. Prove $E(x) \to +\infty$ as $x \to +\infty$ and $E(x) \to 0$ as $x \to -\infty$.

Translate a large-output condition through the inverse.

To show the first limit, let $M > 0$ be any proposed lower bound. The number $L(M)$ is real. Since E is strictly increasing,

$$x > L(M) \quad \Longrightarrow \quad E(x) > E(L(M)) = M.$$

Thus $E(x)$ eventually exceeds every positive bound, which is exactly the definition of divergence to $+\infty$.

Translate a small-output condition in the same way.

Let $\varepsilon > 0$. The range of E is $(0, \infty)$, and

$$x < L(\varepsilon) \quad \Longrightarrow \quad 0 < E(x) < E(L(\varepsilon)) = \varepsilon.$$

This is exactly the condition for $E(x) \to 0$ at $-\infty$.

Conclusion. The two claims follow from monotonicity and the inverse identities; there is no need to assume exponential limit rules.

37.3 Exercise 37.3: All four properties of a general exponential base

What we must show. Prove the assertions of Theorem 37.8 for fixed $b > 0$ and $B(x) = b^x = E(xL(b))$.

(i) Continuity and differentiability.

The inner function $x \mapsto xL(b)$ is linear; its derivative is the constant $L(b)$. Since E is differentiable on all of $\mathbb{R}$, the chain rule gives

$$B'(x) = E'(xL(b))L(b) = L(b)E(xL(b)) = L(b)b^x.$$

Differentiability also implies continuity. This includes $b = 1$: $L(1) = 0$, so $B(x) = E(0) = 1$ and $B' = 0$.

(ii) Determine the direction of monotonicity.

Because L is strictly increasing and $L(1) = 0$, $L(b) > 0$ for $b > 1$, whereas $L(b) < 0$ for $0 < b < 1$. In the first case the linear inner function is strictly increasing, so its composition with increasing E is strictly increasing. In the second case it is strictly decreasing, so the composition is strictly decreasing.

(iii) Verify the range, rather than merely stating it.

Assume $b \neq 1$, so $L(b) \neq 0$. Every value $B(x)$ is positive, because E is positive. Conversely, given $y > 0$, choose

$$x = \frac{L(y)}{L(b)}.$$

Then $B(x) = E(L(y)) = y$. Thus every positive value is reached, and the range is exactly $(0, \infty)$.

(iv) The addition rule in the exponent.

The defining formula and Theorem 37.6(ii) yield

$$b^{u+v} = E((u+v)L(b)) = E(uL(b) + vL(b)) = E(uL(b))E(vL(b)) = b^u b^v.$$

Conclusion. All assertions are proved, with the constant-base case $b = 1$ kept separate where invertibility is required.

37.4 Exercise 37.4: Change of logarithm base

What we must show. For $b > 0$, $b \neq 1$, prove $\log_b y = L(y)/L(b)$ for $y > 0$.

Use the definition of the inverse function.

Let $x = \log_b y$. By definition, $b^x = y$. Since $b^x = E(xL(b))$, applying L to both sides gives

$$xL(b) = L(E(xL(b))) = L(y).$$

The denominator $L(b)$ is nonzero because L is one-to-one and $L(1) = 0$. Division therefore gives $x = L(y)/L(b)$.

Conclusion. $\log_b y = \dfrac{L(y)}{L(b)}$. In particular, since $L(e) = 1$, $\log_e y = L(y)$.

37.5 Exercise 37.5: Differentiate a real power

What we must show. For fixed $p \in \mathbb{R}$, prove that $f(x) = x^p$ on $(0, \infty)$ has derivative px^{p-1}.

Start with the definition of a real power.

For a positive base x, the definition says $f(x) = E(pL(x))$. Both functions in this composition are differentiable on the relevant domains. The chain rule gives

$$f'(x) = E'(pL(x))\frac{p}{x} = \frac{p}{x}E(pL(x)) = \frac{p}{x}x^p.$$

Justify the exponent simplification.

The product rule for E gives $E(t)E(-t) = E(0) = 1$, so $E(-L(x)) = 1/x$. Consequently

$$\frac{x^p}{x} = E(pL(x))E(-L(x)) = E((p-1)L(x)) = x^{p-1}.$$

This also works for $p = 0$, since then the derivative is zero.

Conclusion. $(x^p)' = px^{p-1}$ for every real p and every $x > 0$.

37.6 Exercise 37.6: A common real power of a product

What we must show. Show $x^p y^p = (xy)^p$ when $x, y > 0$ and $p \in \mathbb{R}$.

Rewrite both powers using E and L.

The product law for L and the sum law for E give

$$\begin{aligned}(xy)^p &= E(pL(xy)) \\ &= E(p[L(x) + L(y)]) \\ &= E(pL(x) + pL(y)) \\ &= E(pL(x))E(pL(y)) = x^p y^p.\end{aligned}$$

Each logarithm is defined because the two bases and their product are positive. Multiplication by a negative p causes no problem; no inequality has been multiplied by p.

Conclusion. The identity holds for all real exponents under the stated positive-base assumption.

37.7 Exercise 37.7: Derivatives of exponential and logarithmic functions

(a) Differentiate $B(x) = b^x$.

For $b > 0$, the chain-rule calculation from Exercise 37.3 gives

$$B'(x) = E'(xL(b))L(b) = L(b)b^x = (\log_e b)b^x.$$

When $b = 1$ the formula correctly reads $B' = 0$.

(b) Differentiate the inverse, checking the base restriction.

The logarithm $\log_b$ exists only for $b > 0$, $b \neq 1$. Exercise 37.4 expresses it as a constant multiple of L:

$$\frac{d}{dy}\log_b y = \frac{1}{L(b)}L'(y) = \frac{1}{yL(b)} = \frac{1}{y\log_e b}, \qquad y > 0.$$

For $0 < b < 1$ this derivative is negative, agreeing with the decreasing character of that logarithm.

Conclusion. $(b^x)' = (\log_e b)b^x$ and $(\log_b y)' = 1/(y\log_e b)$ on their stated domains.

37.8 Exercise 37.8: A variable base and a variable exponent

What we must show. Differentiate $f(x) = x^x$ for $x > 0$.

Plan. The fixed-exponent power rule cannot be applied directly, because the exponent now changes with x.

Rewrite before differentiating.

By the definition of real powers, $f(x) = E(xL(x))$. The product rule gives

$$\frac{d}{dx}[xL(x)] = L(x) + x\frac{1}{x} = L(x) + 1.$$

Applying the chain rule and $E' = E$ now yields

$$f'(x) = E(xL(x))[L(x) + 1] = x^x[1 + \log_e x].$$

Conclusion. $(x^x)' = (1 + \log_e x)x^x$ for $x > 0$.

37.9 Exercise 37.9: An elementary growth comparison for the logarithm

What we must show. Prove the requested inequalities and then $\lim_{y\to\infty}(\log_e y)/y = 0$ without L'Hospital's rule.

(a) Bound the defining integral.

For $y > 1$, we have $0 < 1/t \leq 1$ on $[1, y]$. Monotonicity of the ordinary integral gives

$$0 < \log_e y = \int_1^y \frac{dt}{t} \leq \int_1^y 1\,dt = y - 1 < y.$$

(b) Apply the first estimate at a square root.

Since $\sqrt{y} > 1$, part (a) gives $\log_e \sqrt{y} < \sqrt{y}$. The product identity for L gives $\log_e y = \log_e(\sqrt{y}\sqrt{y}) = 2\log_e \sqrt{y}$. Dividing by positive y preserves the inequality:

$$0 < \frac{\log_e y}{y} < \frac{2\sqrt{y}}{y} = \frac{2}{\sqrt{y}}.$$

(c) Turn the estimate into a limit.

The rightmost term tends to zero. More explicitly, if $\varepsilon > 0$ and $y > \max\{1, 4/\varepsilon^2\}$, then $0 < (\log_e y)/y < \varepsilon$. This is precisely the required limit definition.

Conclusion. The logarithm grows more slowly than y: $\lim_{y\to\infty} \dfrac{\log_e y}{y} = 0$.